Predicting Olympic Medal Success: Paris 2024 Project Report

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1. Motivation

Olympic medals are rare. In Paris 2024, only a minority of athletes (roughly 10 to 20%) stood on a podium, yet sports federations, broadcasters, analysts, and fans all want to know who is likely to win and why. This project asks a focused question: **given what is known about an athlete and their event, can we estimate the probability that they win a medal?**

The project is motivated by four goals:

- **Practical forecasting.** Medal forecasts are used by national committees for funding decisions, and by media for coverage planning.
- **Understanding drivers of success.** Do physical attributes, team versus individual events, or national delegation strength matter most?
- **Demonstrating an honest ML workflow.** The medal label is heavily imbalanced, so a careless model can look accurate while being useless. The project shows how to build a baseline, expose its weaknesses, and improve it step by step.
- **A path to real prediction.** The question that matters most is forward-looking: what is an athlete's chance of medaling at the next Games (LA 2028)?

2. Dataset Description

The project uses the Kaggle *Paris 2024 Olympic Summer Games* dataset: a set of linked CSV tables on the athletes, teams, events, schedules, medals, and national medal standings of the 2024 Games, with more than 200 National Olympic Committees (NOCs) represented. The table below lists the main tables and how each is used. Exact file and column names vary by dataset version, so they should be confirmed with df.columns when loading.

Table	One row per	Key fields	Role in the project
athletes	Athlete	Name, gender, NOC code, birth date, height, weight, disciplines, events	Main feature table
medallists / medals	Medal awarded	Athlete, medal type, event, discipline, country, date	Source of the won_medal label
teams	Team entry	Team name, country, discipline, events, members	Team vs. individual flag; grouping for team-aware splits
events	Event	Event name, sport, discipline	Event and discipline context

Table	One row per	Key fields	Role in the project
schedules	Competition session	Date, start and end time, discipline, event, venue, phase	Number of sessions per event
medals_total	Country	Gold, silver, bronze, total	Country-level analysis and forecast; not an athlete-level feature
nocs	Country	NOC code, country name	Maps codes to country names

Unit of analysis and label. The modeling table has one row per athlete. The target won_medal is 1 if the athlete appears in the medals table and 0 otherwise. The true medal rate should be computed from the data before modeling, because it sets the no-skill PR-AUC. It is expected to fall roughly between 10% and 20%.

Data quality issues to handle. Height and weight are missing for many athletes and are imputed by discipline and gender. Athletes who enter several events appear in several schedule rows and must be aggregated. Team events award a medal to every member, so teammates share a label. Country-level tables such as medals_total contain the answer in aggregate and must not be used as athlete-level features.

3. Baseline Model

A baseline sets the floor that every later model must beat. Two baselines are used.

A) Majority-class baseline

A DummyClassifier(strategy='most_frequent') ignores all features and predicts "no medal" for every athlete. Because most athletes (roughly 80 to 90%) did not medal, it reaches an equally high accuracy while identifying **zero** medalists (recall = 0, F1 = 0). This exposes the accuracy trap and is the reason the project evaluates with PR-AUC, F1, and recall on the medal class instead of accuracy.

B) Logistic regression baseline

A logistic regression is trained on a small set of raw features: age, height, weight, gender, discipline, country, and a team-event flag.

- Numeric features are median-imputed and standardized.
- Categorical features are one-hot encoded, ignoring unseen categories.
- Preprocessing and model are wrapped in a single scikit-learn Pipeline so that nothing is fit on test data.
- The split is an 80/20 stratified split with a fixed random seed.

C) Evaluation metrics

Metric	Why it is used
PR-AUC	Summarizes ranking quality on the rare class. The no-skill value equals the medal rate (roughly 0.10 to 0.20), which is the number to beat.
Recall	How many actual medalists the model finds.
Precision	How many predicted medalists are correct.
F1	Balance of precision and recall.
Accuracy	Reported only to show why it is misleading here.

4. Gaps in the Baseline

The baseline is deliberately simple, and its weaknesses define the rest of the project.

1. **Accuracy is misleading.** The majority-class model scores 80 to 90% accuracy and finds no medalists. The logistic baseline has similar accuracy and likely low recall.
2. **Too little information.** Raw age, height, and weight say little on their own. A 190 cm athlete is tall in gymnastics and average in basketball. The model lacks context.
3. **Class imbalance.** With only 10 to 20% positives, the model can lean toward "no medal" and rarely predicts the minority class.
4. **Linearity.** Logistic regression cannot capture interactions (for example, tall and in a team event and from a large delegation) or non-linear effects such as an age sweet spot in some sports.
5. **Default settings and a single split.** One train/test split can give a lucky or unlucky score, and default hyperparameters are rarely optimal.
6. **Arbitrary decision threshold.** The default 0.5 cutoff is not suitable for imbalanced data.
7. **Data leakage risks.** Country medal counts, or any feature derived from the medal tables, would leak the label. In team events, one medal marks every member as a medalist, so near-duplicate rows can appear in both train and test sets.
8. **No past performance.** A single-Games dataset contains no athlete history, and prior results are the strongest real-world predictor of a medal. This is the most important limitation. It makes the baseline a retrospective model that explains 2024 outcomes rather than a true forecaster.

5. How We Improve on These Gaps

Improvements are added one at a time and logged in a comparison table, so each gain can be attributed to a specific change.

Step	Gap addressed	Change	Metric expected to move
1. Feature engineering	Too little information	Age at the Games, BMI, height and weight relative to discipline average, number of events, team flag, discipline size, delegation size	PR-AUC, recall
2. Imbalance handling	Minority class ignored	class_weight='balanced', SMOTE (training folds only), or threshold tuning	Recall, F1
3. Random Forest	Linearity	Captures interactions and non-linear effects automatically	PR-AUC
4. XGBoost / LightGBM	Linearity, hard cases	Sequential boosting focuses on borderline athletes and handles imbalance natively	PR-AUC
5. Hyperparameter tuning with stratified k-fold CV	Defaults, single split	RandomizedSearchCV or Optuna; report CV mean and standard deviation	PR-AUC, stability
6. Threshold optimization	Arbitrary cutoff	Choose the threshold on the precision-recall curve using validation data	F1, precision/ recall balance

A) Guarding against leakage

- Do not use features derived from medal tables, such as a country's total medals.
- Fit any target encoding inside cross-validation folds only.
- Use GroupKFold by team so that members of a team medal do not appear in both train and test.

B) Interpretation and error analysis

SHAP values and feature importances show what drives medal success, for example discipline and team-event status versus physical attributes. Per-discipline error analysis shows where the model fails, which tells us where more data or better features are needed.5.3 Results table (to be completed)

Model	PR-AUC	Recall	Precision	F1
Majority class				
Logistic regression (baseline)				
+ engineered features				
+ imbalance handling				
Random Forest				
XGBoost / LightGBM				

Model	PR-AUC	Recall	Precision	F1
Tuned model				
Tuned model + optimized threshold				

Not every step is guaranteed to help. If a step adds nothing, that is a legitimate finding and will be reported as such.

6. Closing the Biggest Gap: From Explaining to Forecasting

Gaps 1 to 7 are addressed by the steps above. Gap 8, the lack of history, requires a change in data design.

1. **Build a panel dataset**: one row per athlete per Games, with features computed only from information available before that Games (previous medals, best result in the last cycle, years of experience, country medal history in the discipline).
2. **Split by time, not at random**: train on earlier Games, validate on Paris 2024 as a genuine out-of-sample test, then retrain through 2024.
3. **Score current athletes for LA 2028** using the same feature-building function, with age and history computed as of 2028.
4. **Calibrate** probabilities (calibration curve, Brier score, isotonic or Platt scaling) so that a reported 30% really means about 30%.

Stronger sources, such as World Championships results, world rankings, and qualifying marks, would improve accuracy substantially but must be collected from federation sites.

Expect modest accuracy at the athlete level, since country totals are far more predictable, and note that LA 2028's new sports will have no history to learn from.

7. Future Use

- **LA 2028 medal watch list.** A ranked list of current athletes by estimated medal probability, interpreted as P(medal | the athlete competes), since injuries, retirements, and qualification are not modeled.
- **Country-level forecasts.** Aggregating athlete probabilities gives expected medals by country and discipline, which is the more reliable and defensible output.
- **Resource allocation.** Federations could use the model to see which disciplines or athletes sit near the medal boundary and may benefit most from support.
- **Talent identification.** Feature importance can highlight which attributes and experience profiles are associated with success in each discipline.
- **Reusable pipeline.** The same pipeline (panel features, time-based validation, calibration) applies to other competitions, such as the Winter Olympics, World Championships, or professional leagues.
- **Dashboard.** Predictions and explanations could be served through a small web application, for example Flask with pandas, so users can explore athletes and countries interactively.

8. Conclusion

This project starts from a deliberately simple baseline that shows why accuracy is the wrong yardstick on imbalanced data, identifies its concrete gaps, and closes them one at a time with measurable improvements. Its most important contribution is the move from a retrospective model trained on one Games to a time-aware forecasting pipeline that can score the next one.